

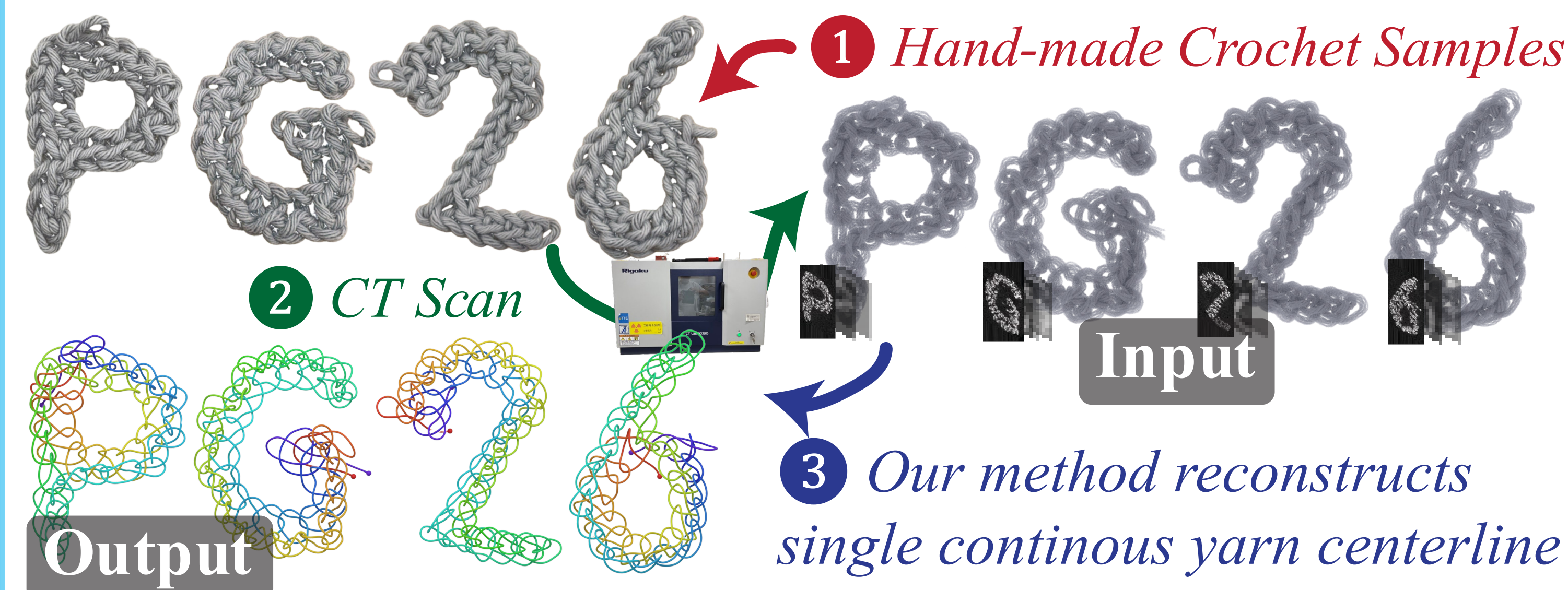
CT2Yarn: Yarn Level Reconstruction of Crochet from Computed Tomography

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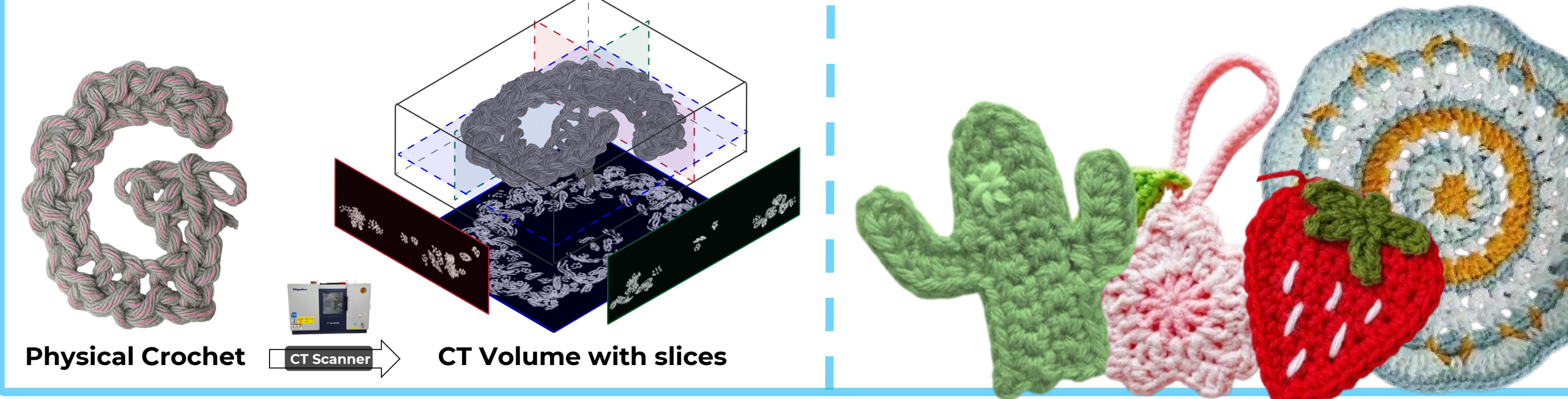
The University of Tokyo

1. Introduction

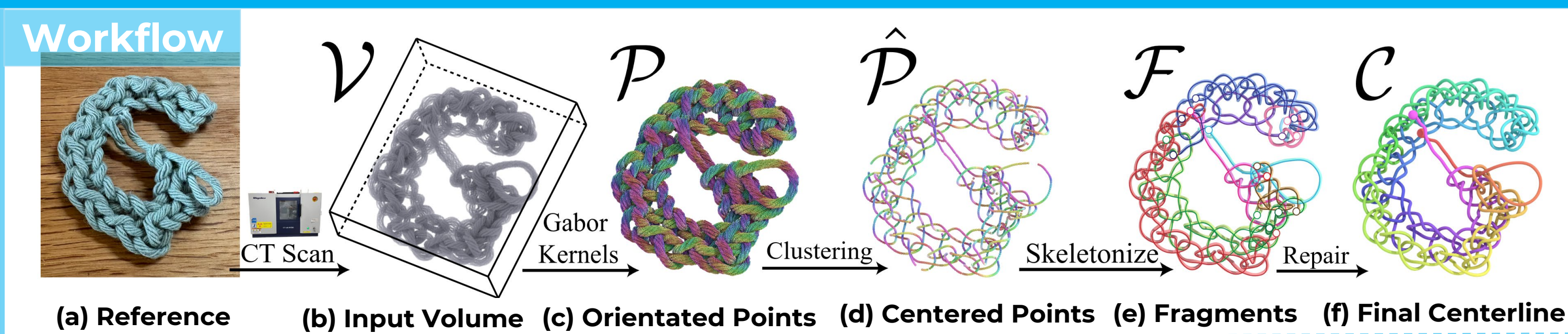
- Human-in-the-loop** method recovers crochet centerline.



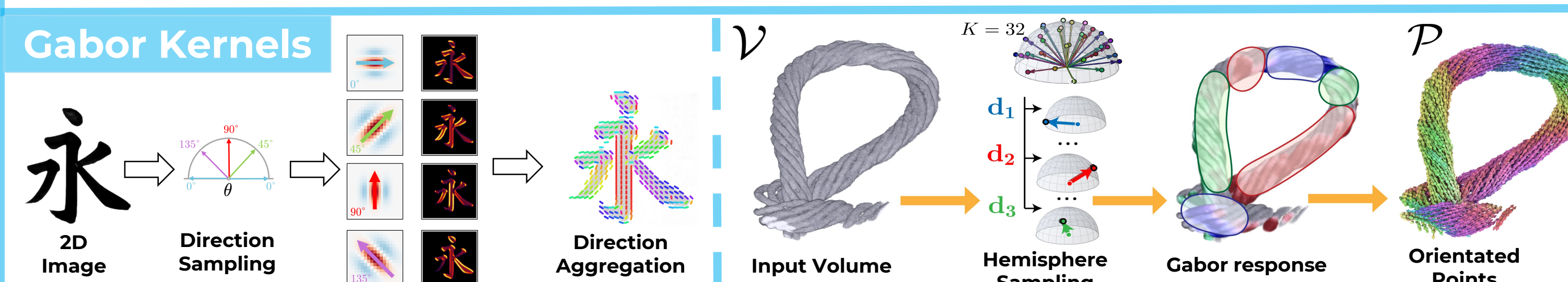
- Micro-CT** reveals the hidden internal structure.



3. Methodology



- Orientation estimation** Gabor extracts local fiber direction.
- Center-point extraction.** Anisotropic mean-shift aggregates directions at the yarn scale. Perpendicular projection, direction-aware weighting, and ellipsoid kernels concentrate points toward the yarn centerline.
- Topology reconstruction.** MST connects fragments. Topology repair resolve common breaks with UI repair.

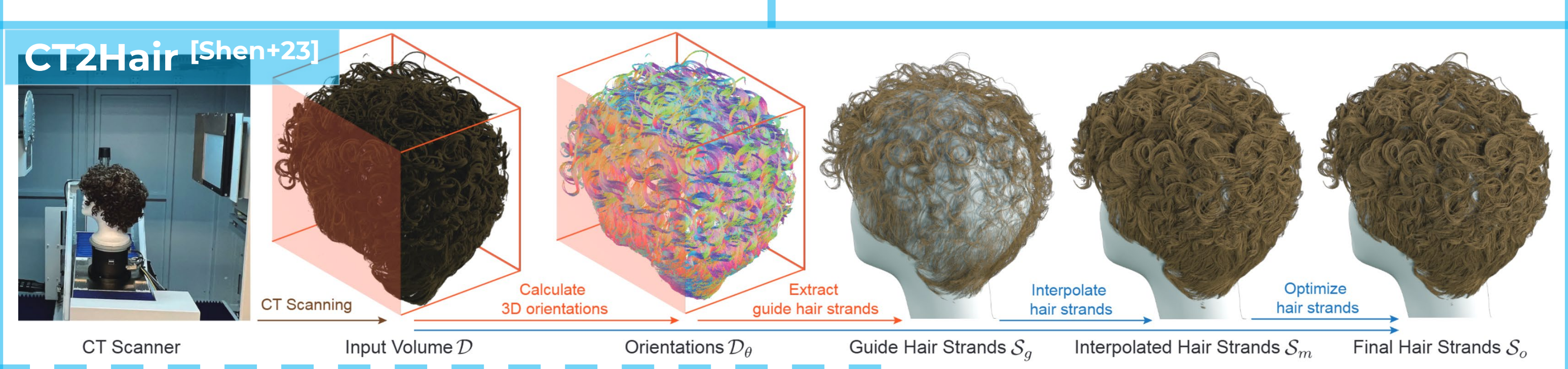
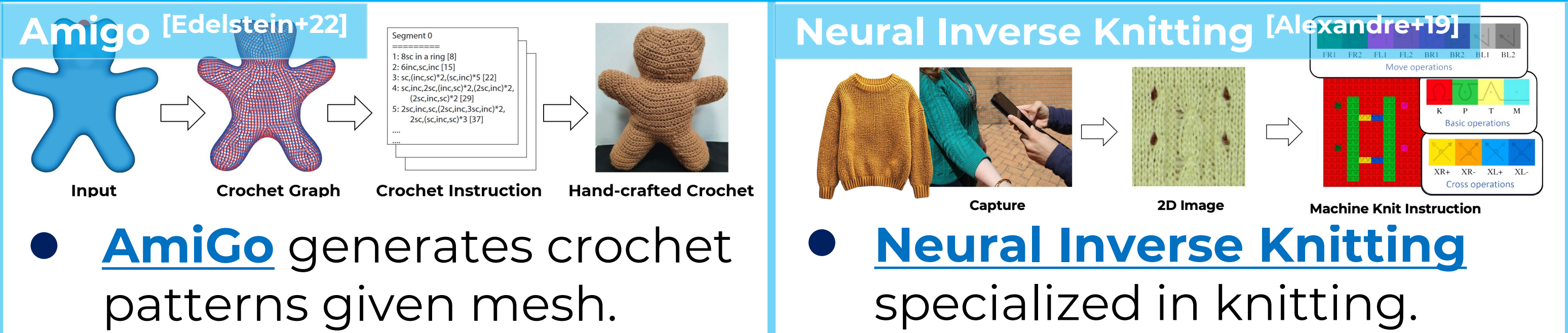


4. Results

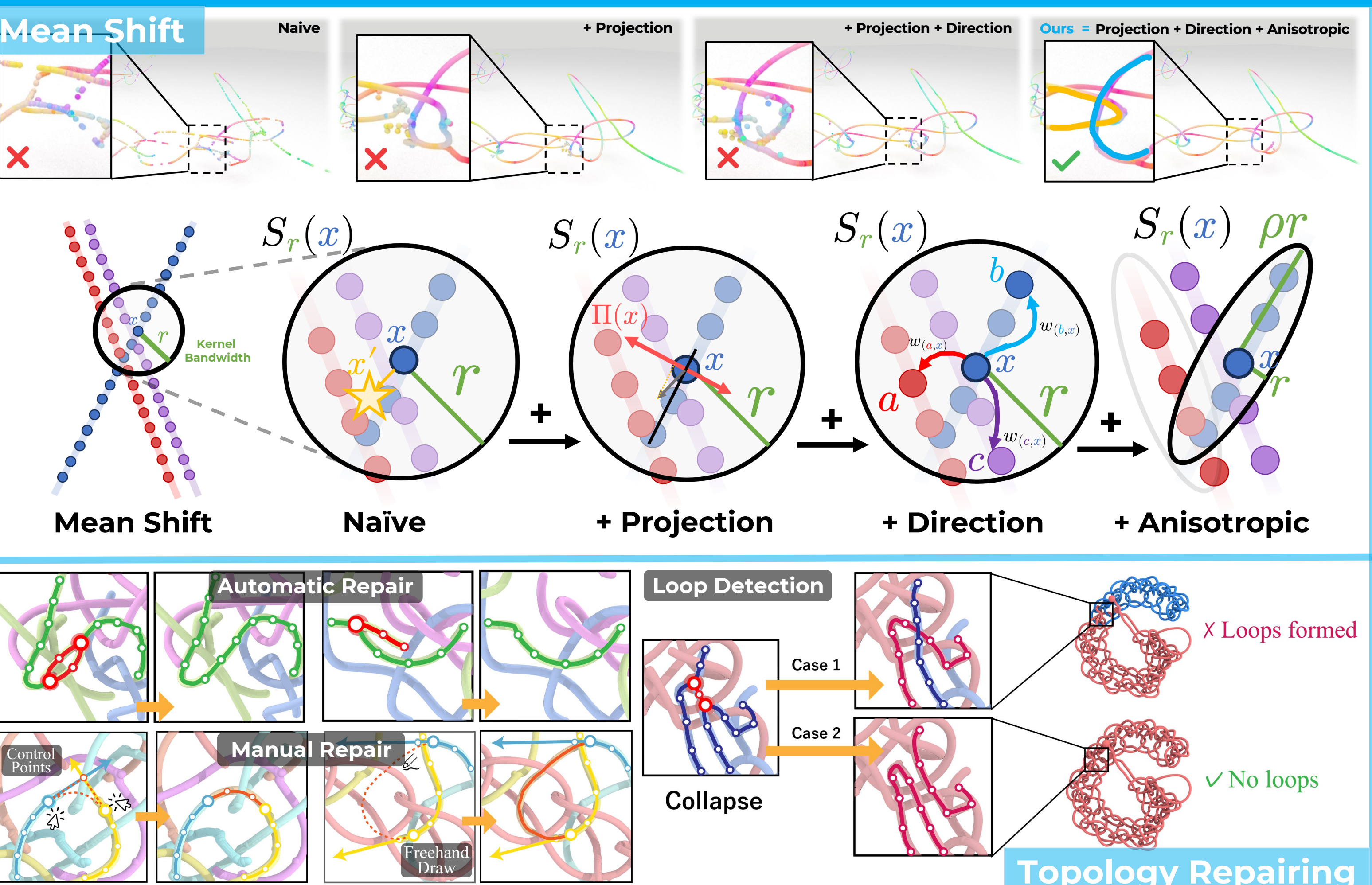
Quantitative Evaluation		Reference-free		Reference-based		Effort	
Method	#C	$\bar{L}_C$	$E_L \downarrow$	$\bar{\alpha} \downarrow$	$\ell_{CD} \downarrow$	$\ell_2 \downarrow$	edits time
CT2Hair (Gab.)	88	314	0.115	10.55	16.99	18.55	— —
CT2Hair (Grad.)	115	236	0.271	12.75	20.60	24.67	— —
Ours (Auto)	15	3814	0.002	2.66	2.43	2.26	— —
Ours (Manual)	1	30359	—	—	—	—	4 2:46



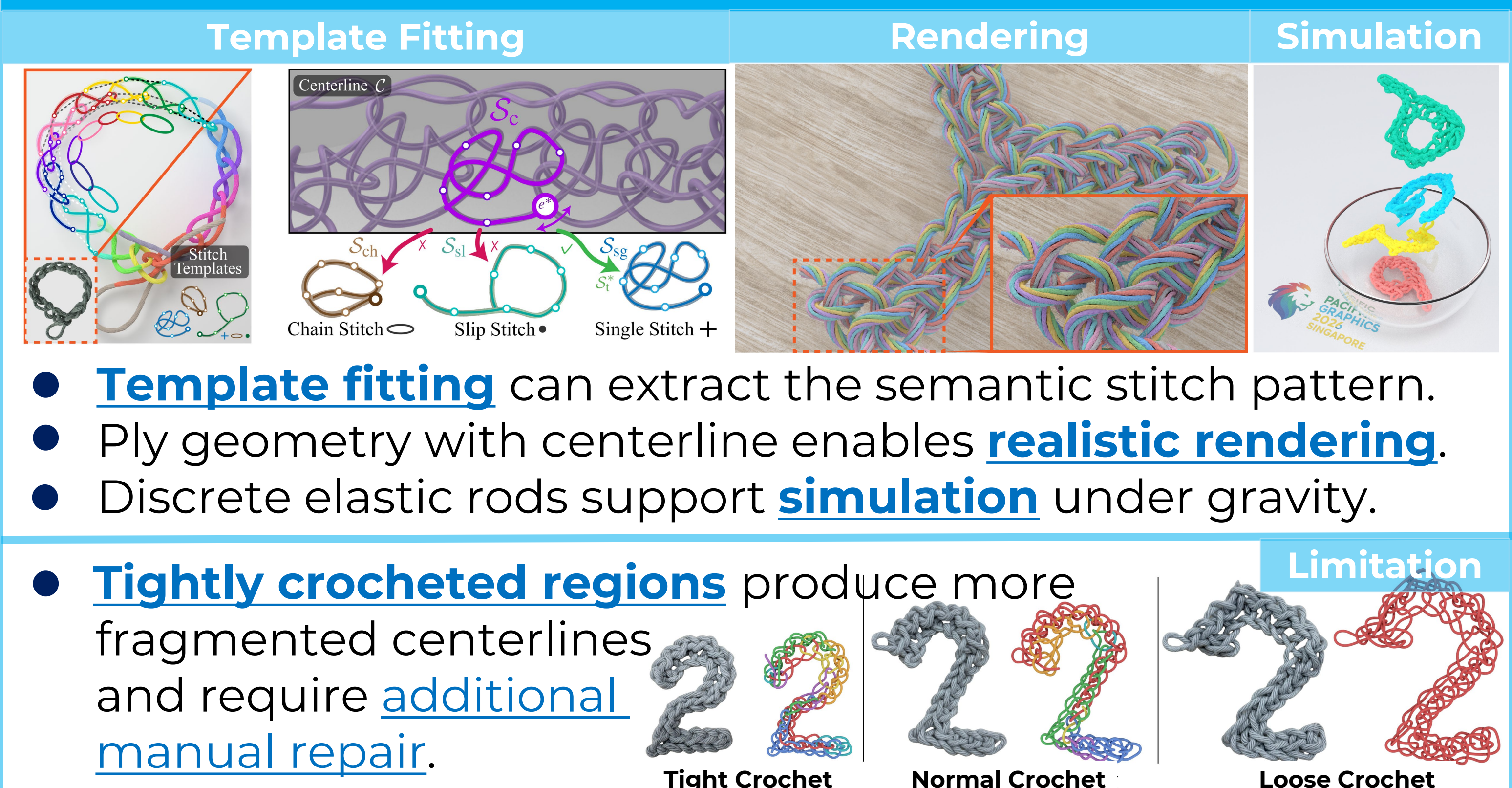
2. Related works



- CT2Hair** reconstruct hair strands from wig CT scan.
- Crochet has **Hierarchy!**
- Wrong connections in local** brings error in **global topology!**



5. Applications and Limitations



- This work was supported by **JST SPRING**, Grant Number JPMJSP2108.

- References:**
 - Shen, Y., Saito, S., Wang, Z., Maury, O., Wu, C., Hodgins, J., ... & Nam, G. (2023). Ct2hair: High-fidelity 3d hair modeling using computed tomography. *ACM Transactions on Graphics (TOG)*, 42(4), 1-13.
 - Edelstein, M., Peleg, H., Itzhaky, S., & Ben-Chen, M. (2022, October). Amigo: Computational design of amigurumi crochet patterns. In *Proceedings of the 7th Annual ACM Symposium on Computational Fabrication* (pp. 1-11).

